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Data Science Fundamentals

Hierarchical Clustering

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Before we get started…

Practicalities

Learning objectives

- You understand and apply different types of regression, know when and how to use them, can execute them and put them in to practice as well as interpret the results.
- You recognize overfitting in regression models and learn techniques to prevent overfitting.
- You learn about different clustering techniques and understand when and how to use them for grouping data.
- You gain an understanding of dimensionality reduction techniques and learn how to apply them to compress and visualize data.

Course material

- **Canvas**
  - Slides & exercises from classes
- **Online**
  - https://www.w3schools.com/datascience/
- **Software**
  - JupyterLab (or work via Anaconda/VS Code if you prefer)
- **The Internet**



Course breakdown

- **Course Structure:**
  - Lectures: 3 hours per week (theory and exercises)
  - Attendance: Mostly not compulsory, but strongly recommended.

- **Examination Breakdown:**
  - 30%: Continuous assessment (mandatory exercises)
  - 70%: Final exam (oral, mostly theoretical with some practical elements)

Is it difficult to pass?

It’s perfectly doable **if you put in the work**

- Come to class
- Participate in the lessons
- Make your exercises



How to be an ideal student

- **Attend classes** - even when you party hardy
- **Show respect** - towards your instructor and fellow students
- **Be punctual** - or you'll find the door closed
- **Meet deadlines** - late submissions will earn a solid zero
- **Have a question?** - just ask! I don't bite
- **Know it all?** - help your classmates, they might repay you with pizza
- **Do your own work!** - shortcuts will lead to a short academic career ;)

Syllabus

Week - 1	Introduction & Python/JupyterLab Setup	Week - 9	Easter Holiday
Week - 2	Regression - Linear regression	Week - 10	Easter Holiday
Week - 3	Regression - Polynomial regression	Week - 11	Clustering - Hierarchical clustering
Week - 4	Classification - Logistic regression	Week - 12	Dimensionality reduction: Feature Analysis - PCA
Week - 5	Classification - Decision trees	Week - 13	Regularization: Lasso regression
Week - 6	Classification - kNN	Week - 14	Recap: Theory + Excercises
Week - 7	Clustering - kMeans	Week - 15	Exam Preperation + Q&A
Week - 8	Lifesnaps guidance		

(subject to change)

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Now we get to the fun part

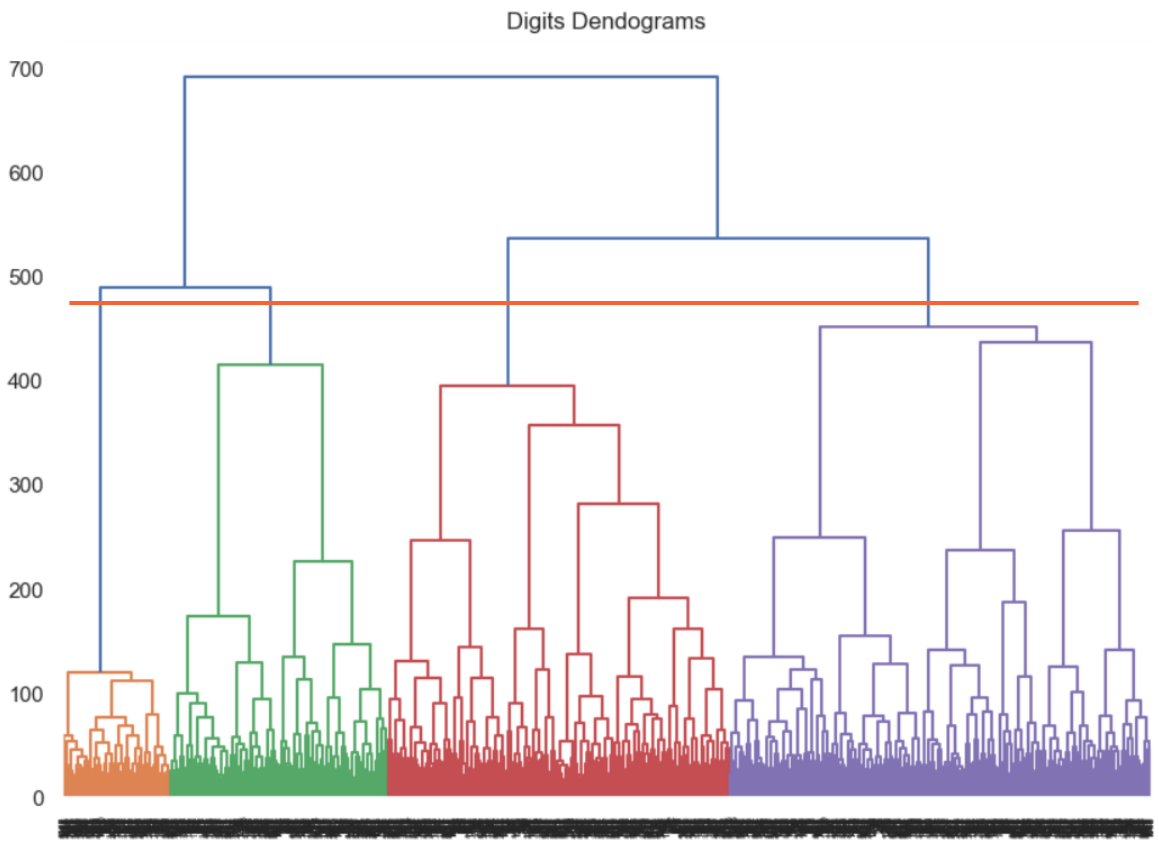
Hierarchical Clustering

Hierarchical Clustering

- More advanced clustering method
- Flexible approach to choosing the amount of clusters
- **Bottom-up** mechanic
  - each individual point starts off as a unique cluster, **merge** until there is only 1 cluster left.
- Clusters are formed by merging the clusters which are **closest together**.

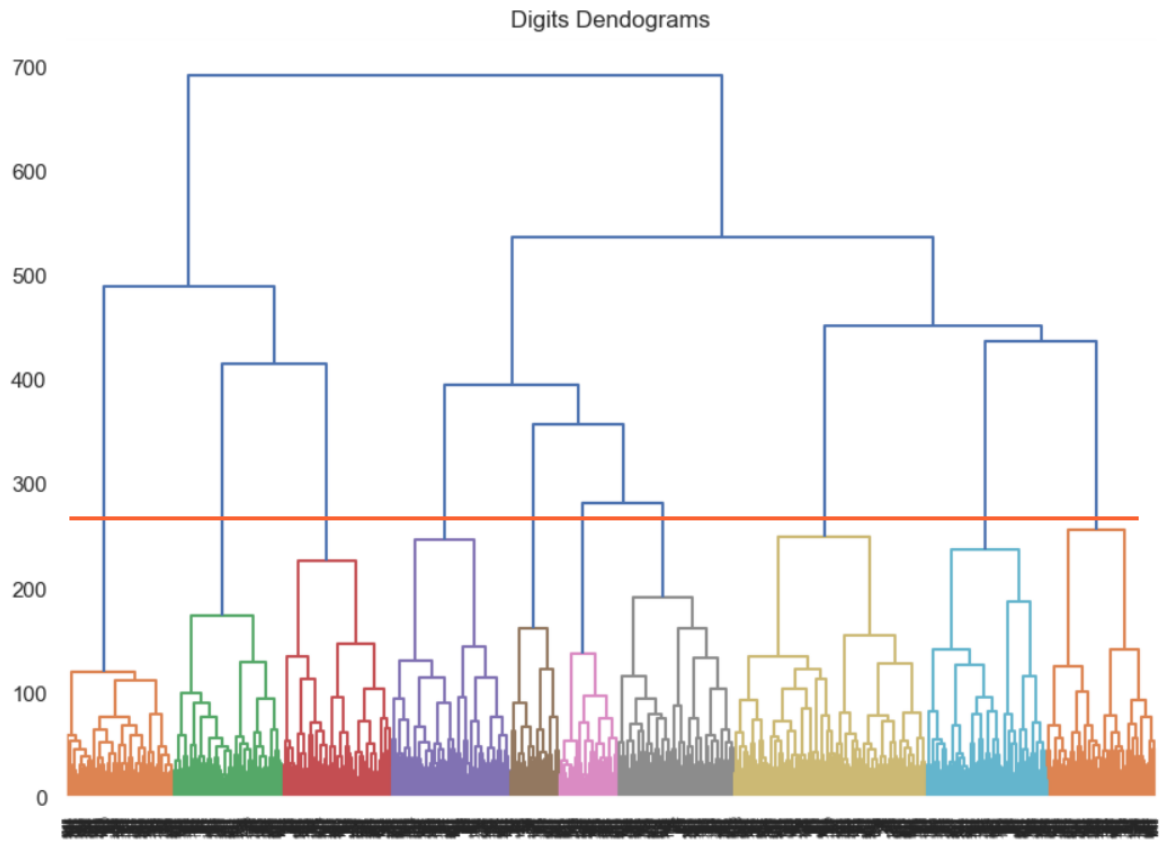
Dendrogram

- Visual representation of **all** cluster combinations
  - From max to 1
- X-axis show the **datapoints**
- Y-axis shows the **distance** between clusters (see Linkage Criterion)
- Lines show the grouping of **clusters**
- Define colours for each cluster (optional but nice :D)



Dendrogram

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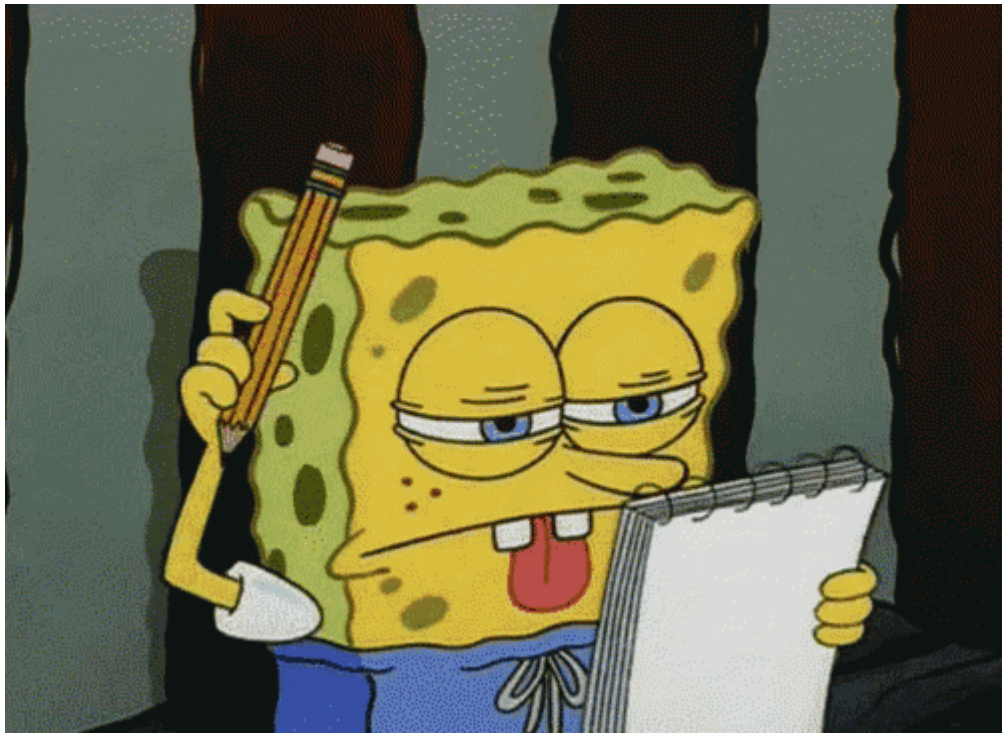


Linkage Criterion

- A way to figure out how ‘good’ a cluster is
- Exact inner workings **differ per model**
- Most common one: ‘**ward**’
- Calculates **inertia** (= inner cluster variance, how close are datapoints within a cluster) for **each** cluster pair
- Similar workings as kmeans → good for **spherical data**
- Dendrogram: y axis plots **inertia**

Finding the best cut-off point

- Draw a **horizontal line** as a cut-off point
- Decided how many clusters we have
- How do we figure out the best value?
- In kmeans we used distortion, however we don’t have this metric here…
- We use what we have: **linkage criterion**!

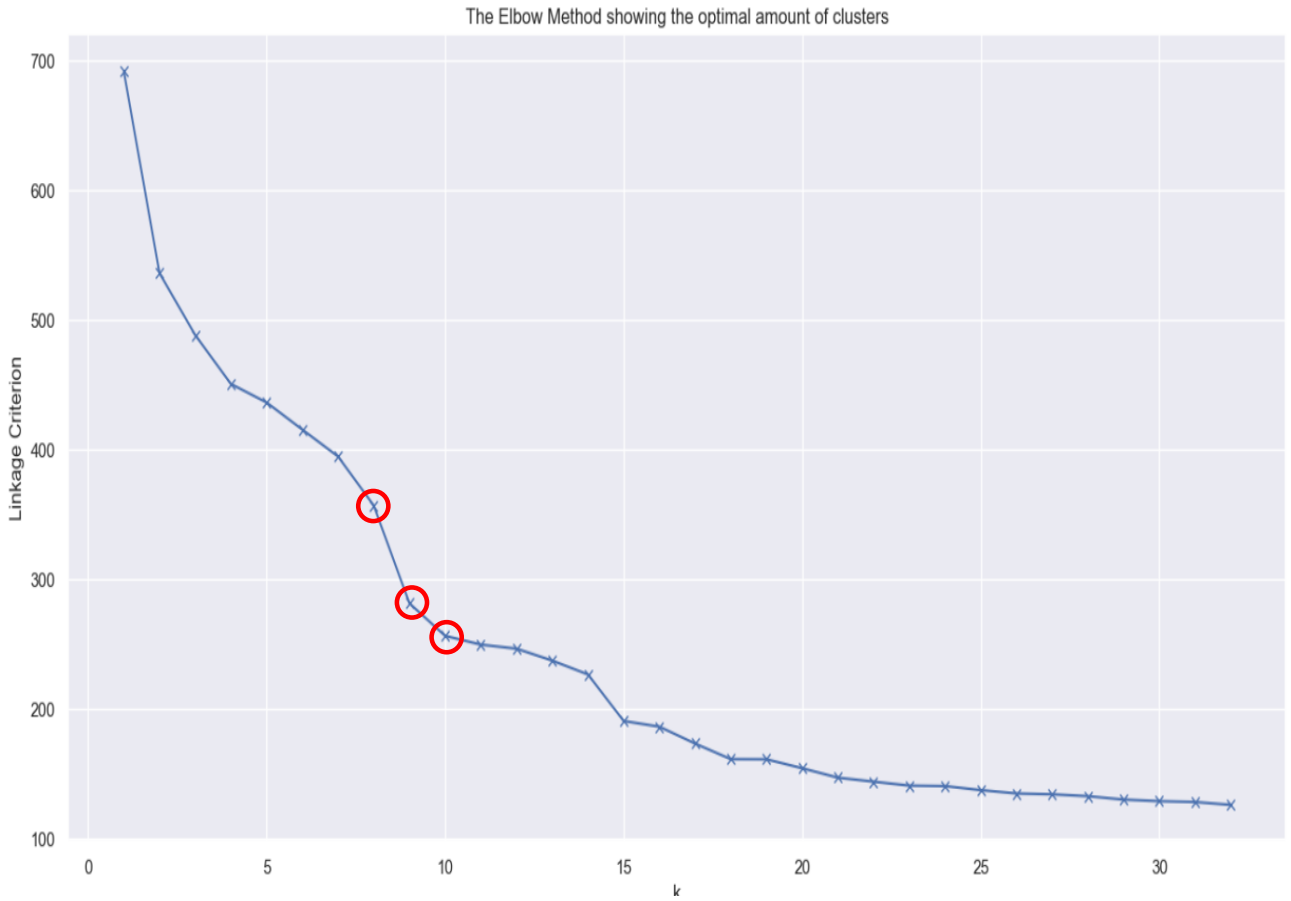


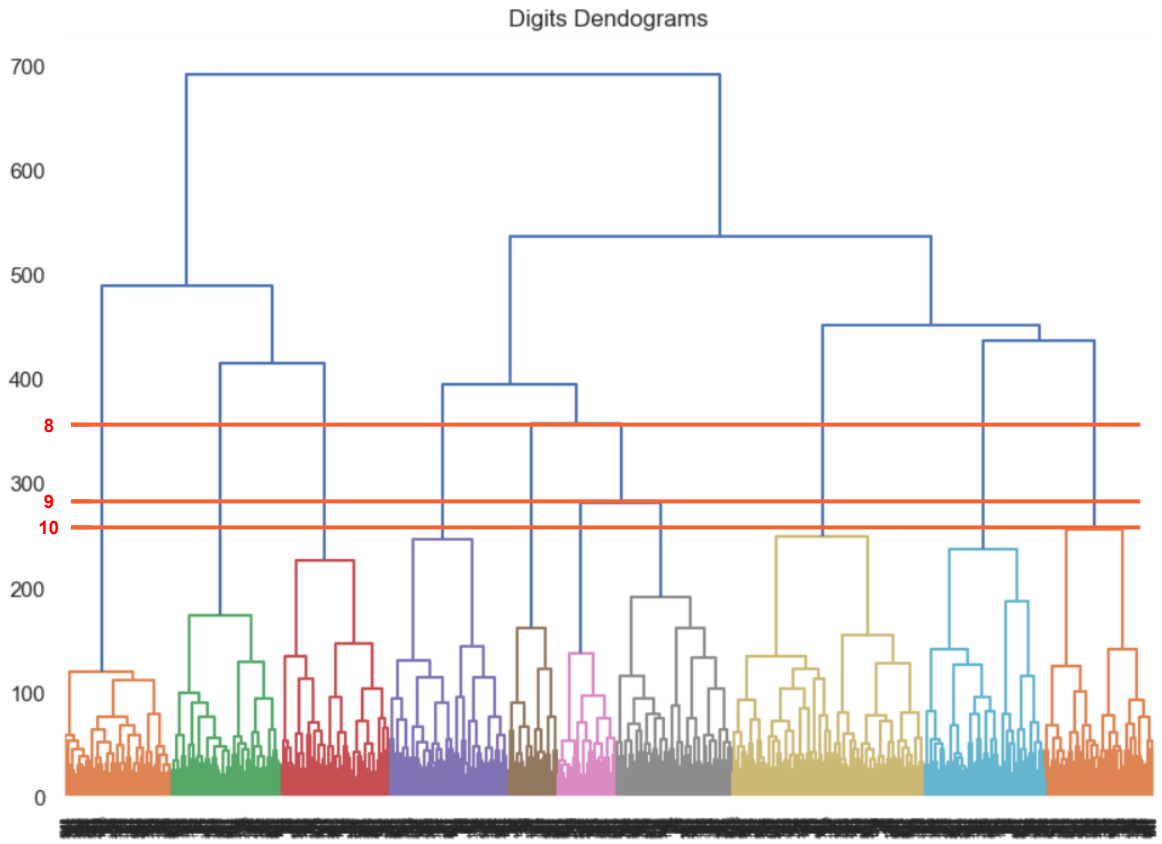
Finding the best cut-off point

- Start with max amount of clusters
  - Ward method: calculate **inertia** for each cluster pair
  - **Merge** the two clusters with the **lowest inertia** = datapoints are closest together
- Repeat until only one cluster remains
- Reverse the array
- Voila, we have our elbow!



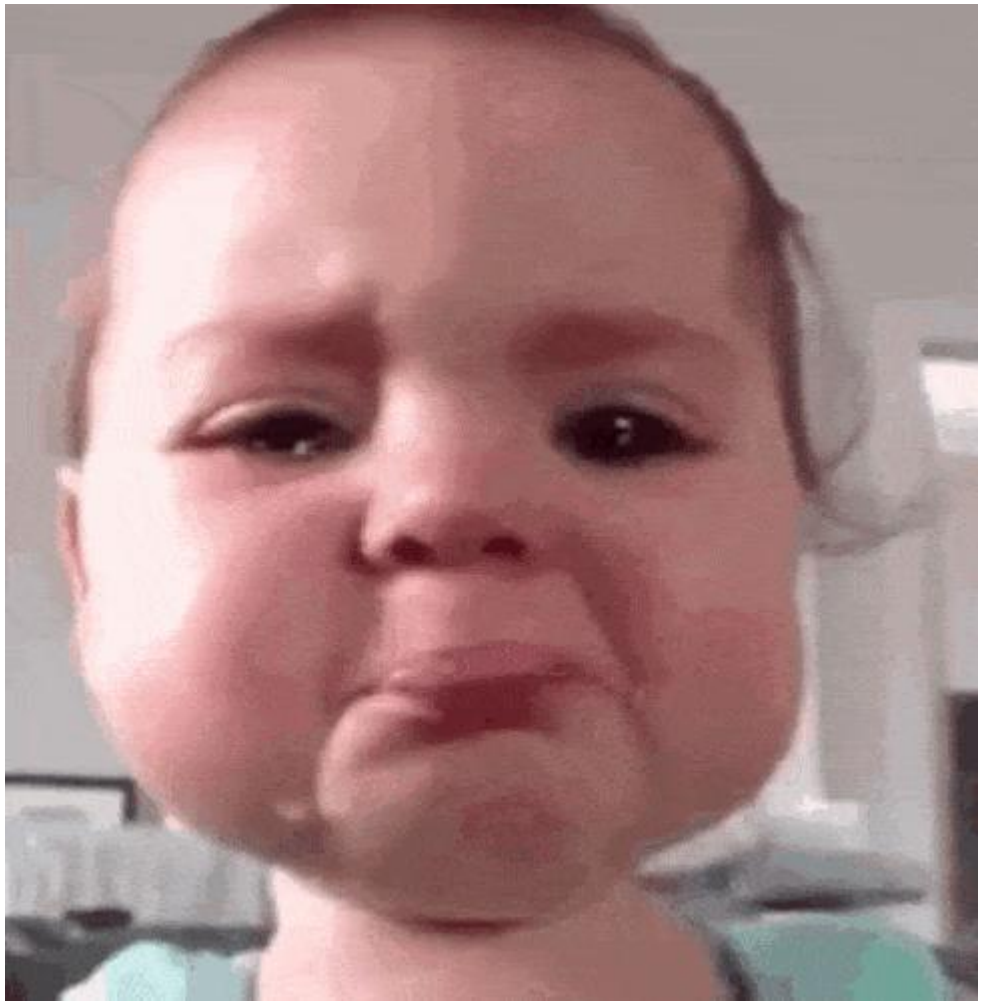
Finding the best cut-off point





Overfitting

- As always, we still have to take into account overfitting
- Overfitting in clustering: **too many clusters**
- Luckily, just like kmeans, this is easy to counter
- Use the elbow method!



Evaluations

- Once again: in clustering we have **no labels**, so there is no way to evaluate our model
- You can plot a **dendrogram** to spot overfitting
- Also once again: we can use our **molding technique** to use clustering to make predictions

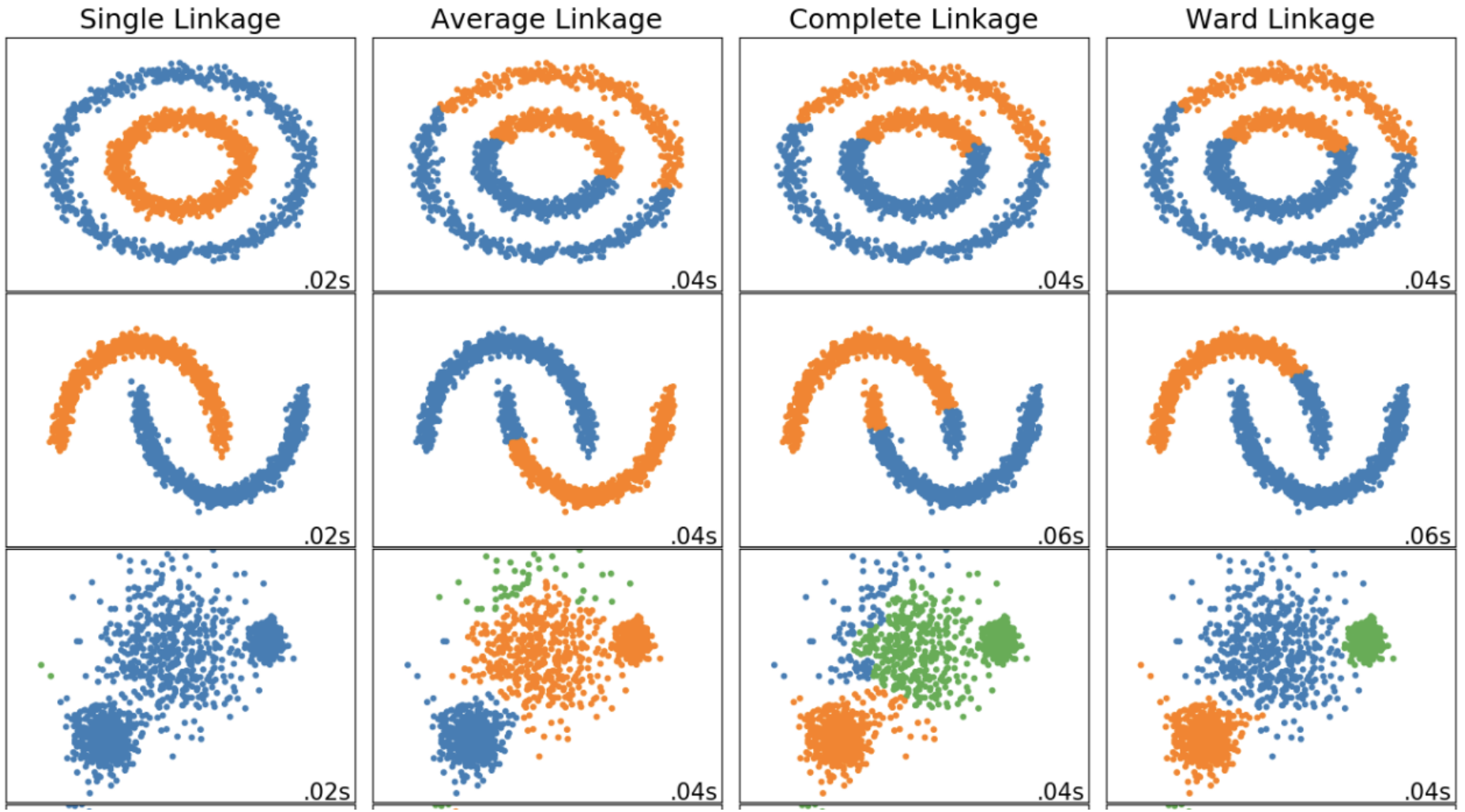


Different mechanics – different dataset

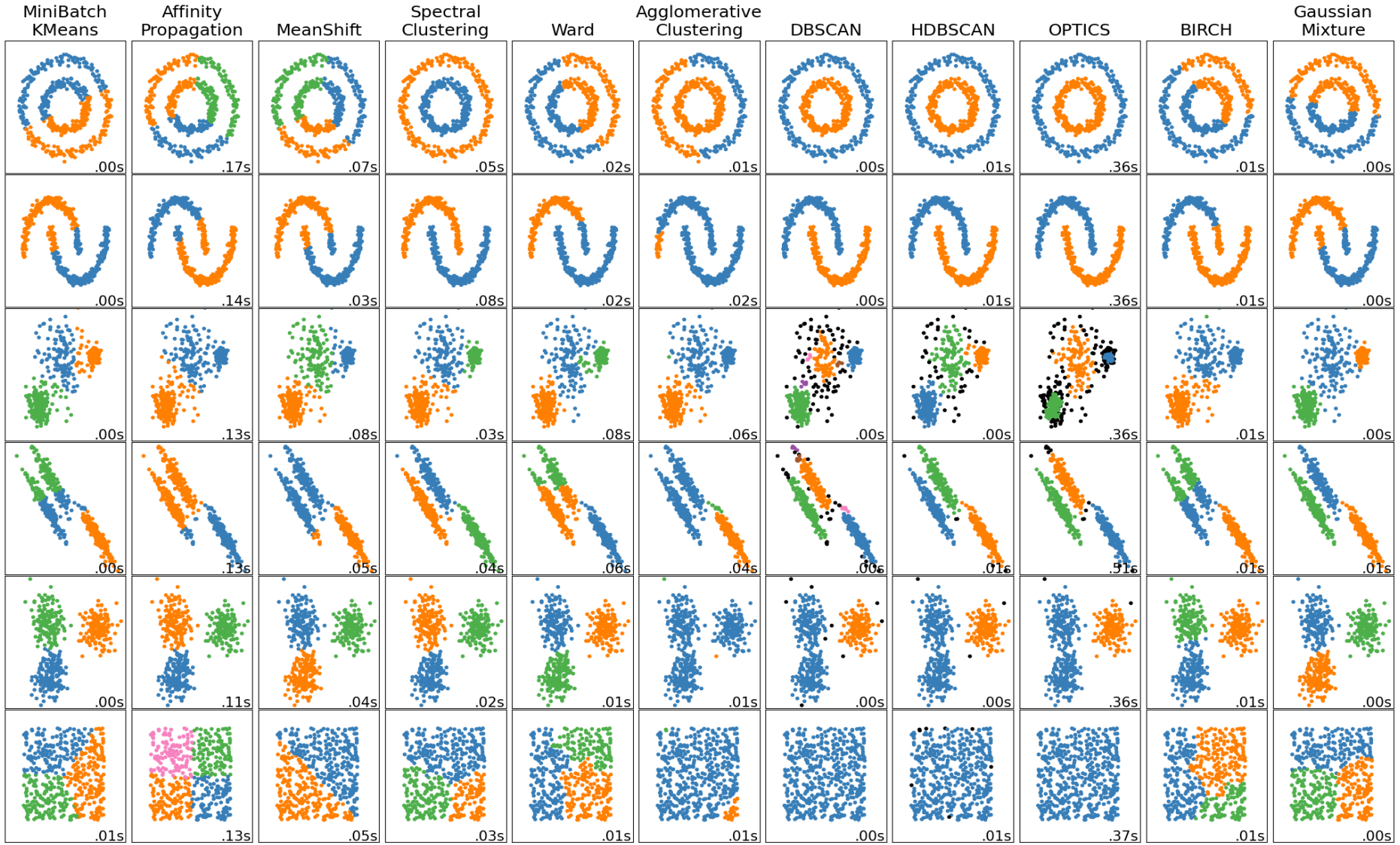
- Remember that I said the ‘ward’ method is the most common one?
- Ward is optimized for **circular clusters**: most common in datasets
- **Other mechanics** are optimized for **different datasets**
- Make sure to **explore** some options!



Different mechanics – different dataset



Other clustering methods



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Let's jump into action

Excercises

First, let's explore this together

- Go to canvas and download the notebook "DSF 08 – Hierarchical Clustering Notebook"
- Open the notebook in Jupyter
  - Open a CMD and run "jupyter lab"
  - Or launch it via anaconda
- Follow along!



Now it's up to you!

- Go to canvas and download the notebook "DSF 08 – Hierarchical Clustering Exercise"
- Also download the csv file called "Cars.csv"
- Place them in the same directory
- Open the notebook in Jupyter
  - Open a CMD and run "jupyter lab"
  - Or launch it via anaconda
- Go through the notebook and make the excercises

I don't bite, ask me any questions you have!

